2026 HiMCM — Problem A

Choosing a City for High-School Life with Incomplete Evidence

Summary Sheet

A city recommendation becomes more useful when its evidence and priorities can be examined together. We compare sixteen cities using ten indicators of safety, environment, household costs, education and everyday resources. Zurich is the conservative recommendation under the retained team inputs and baseline or equal weights. All sixteen cities remain possible champions within the retained evidence bounds.

The evaluation combines fixed-anchor TOPSIS with a maximum-regret decision. Benefit and cost utilities share fixed reference scales, allowing controlled comparisons across data, preference and development scenarios. Missing values remain intervals. We select the city with the smallest worst-case shortfall from a better-scoring competitor over the declared rectangular uncertainty set.

Official municipal records, school statistics and operator inventories supplement twenty-two previously missing entries while preserving the team's original values. The average score-interval width falls from 18.81077 to 16.06144, a 14.62% reduction. The resulting 160-entry matrix contains 112 point values, 34 partial intervals and 14 fully missing intervals. Actual reference years and source populations remain explicit.

A controlled formulation revision retains the 48 uncertain entries rather than collapsing each to a midpoint. Although Zurich remains the recommendation, 13 fixed-weight decision ranks change. Separate interior completions let each of the sixteen cities rank first. Withholding Zurich's unverified homicide-rate point alone selects Singapore under baseline weights and Oslo under equal weights, exposing dependence on accepted source information.

Preferences and development conditions answer different decision questions. Doubling the relative emphasis on education changes the conservative choice to Oslo. In 10,000 baseline weight perturbations at Dirichlet concentration 100, Zurich is recommended in 43.33% of draws, compared with 27.92% for Oslo and 28.23% for Singapore. These are conditional simulation frequencies. Three uniform ten-per-cent development scenarios retain Zurich; separate city-specific shocks reveal conditional recommendation changes without claiming an empirical forecast.

Separate country, district and age-group cases examine what citywide totals leave unresolved. They distinguish facility supply from student access and observed broad age groups from unsupported age-specific preferences. The result is a practical starting choice with explicit alternatives and clear priorities for additional evidence.

Keywords: high-school opportunities; TOPSIS; interval evidence; maximum regret; preference sensitivity.

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1 Introduction and decision scope

1.1 What makes a city suitable for high-school life?

A useful comparison connects a student's school day with the opportunities and constraints around it. Safety and air quality affect the daily setting; teaching resources and nearby higher education describe educational supply; housing, living costs, transport, travel, broadband and swimming facilities describe household pressures and everyday resources. Our operational definition of "best" is the conservative choice under these ten criteria, declared weights and retained evidence ranges. It provides a starting recommendation for the compared cities, rather than a claim about every student's experience.

The study uses the sixteen cities in the frozen team matrix: Amsterdam, Berlin, Dublin, Helsinki, London, Madrid, Milan, Oslo, Paris, Prague, Singapore, Stockholm, Tokyo, Vienna, Warsaw and Zurich. This is an inherited comparison set across sixteen countries, not a representative sample of all world cities. It exceeds the problem's minimum of ten cities and spans different resource and cost settings. The findings concern this set; a broader claim would require an explicit sampling design and additional comparable observations.

1.2 Criterion selection and contribution

The ten indicators preserve the agreed team framework and balance education with the conditions that make resources usable in daily life. Their directions and anchors appear in Table 1. Counts and city averages serve as scoped supply or setting measures. They do not directly measure friendships, student wellbeing, school quality, language fit, discrimination, cultural participation or individual access. Those outcomes are relevant to a family's choice; comparable student-level observations are needed to integrate them without inventing measurements or double-counting existing proxies.

Our central contribution is a city decision that retains incomplete evidence. We improve source precision, compare a point-completion benchmark with an interval revision, and examine preferences and future conditions without changing the primary indicator set. TOPSIS provides one main quality score; AHP structures stated priorities, Monte Carlo measures specified perturbations, and VIKOR supplies a focused method comparison. The extensions test the recommendation rather than vote among an expanding set of ranking models.

AI assistance supported research retrieval, calculations, code and prose preparation, including Exa source retrieval and extraction [18, 19]. The appended use report records the known assistance and the outstanding documentation required before submission.

2 Assumptions and their role

1. **Comparability follows the retained scope.** Original values and population denominators remain fixed. Each supplement keeps its provider, year, city boundary and educational population. Mixed-year densities are conditional comparisons; they do not create a same-year census.

2. **Utilities are monotone on fixed scales.** More resources and less cost or burden improve their respective utilities. Declared anchors set the scale; they are neither estimated preferences nor empirical optimum values.

3. **The score permits compensation.** Weighted distances combine criteria in a common utility space. The model therefore allows stronger results on some criteria to offset weaker results on others. A family requiring a hard safety or eligibility threshold should apply it separately before choosing.

4. **The evidence set is rectangular.** Uncertain utilities vary independently within retained intervals. This conservative closure omits unknown cross-criterion and cross-city restrictions. Its possible champions and regret bounds are conditional on this set; they are not population probabilities.

5. **Extensions hold the comparison basis fixed.** Scenario changes retain population denominators, anchors and the indicated weight rule. They are conditional experiments with no fitted future date or representative age-specific preference distribution.

3 Notation

Symbol	Meaning
M, N	Sixteen cities and ten indicators
w_j, a_j	Normalized criterion weight and fixed utility anchor
u_{ij}	Utility of indicator j for city i, on $[0, 1]$
ℓ_{ij}, h_{ij}	Retained lower and upper utility bounds
L_i, U_i	Lower and upper TOPSIS score, on $[0, 100]$
$R_i, \mathcal{C}$	Maximum regret and set of possible champions
κ	Concentration of a declared Dirichlet weight distribution
r	Raw-value multiplier in a development scenario

4 Data collection and comparability

4.1 A traceable city–indicator matrix

We compare $M = 16$ cities on $N = 10$ indicators covering safety, environment, household costs, education and everyday resources. The initial matrix is the team-supplied city table [1]. We retain its existing values, populations, reference years and source flags as the agreed input, then supplement only empty cells. Some supplied values lack original source URLs, so their provenance remains the team table; official supplements do not retrospectively validate every supplied measurement. Table 1 states what each variable measures.

The nominal reference year is 2025, but the evidence is mixed-vintage. Every supplement retains its actual year rather than being relabelled as 2025. Population denominators remain those of the supplied table. Thus a 2026 facility snapshot divided by a supplied 2025 population is a conditional density, not an observed same-year census. We do not infer that facilities are unchanged merely because they may change slowly. Historical inventory assumptions are evaluated through withdrawal scenarios.

Table 1: Retained indicators, units and reference anchors. Supplied values are retained; supplements have explicit source scope.

Indicator	Measurement and interpretation	Anchor
Homicide	Deaths per 100,000; cost; supplied city proxy	2
$PM_{2.5}$	$\mu g/m^3$; cost; supplied concentration proxy	15
Living costs	Supplied living-cost index; cost	100
Housing	Supplied price-to-income ratio; cost	10
School resources	Pupils per teacher or explicitly scoped FTE; cost	15
Higher education	Institutions per 100,000; benefit; supply proxy	1
Transport stops	Unique stops per 100,000; benefit; partial inventories	50
Travel	Duration in minutes; $\exp(-t/30)$; commute or stage proxy	30 min
Broadband	Supplied Mbps; benefit; city/general-use proxy	100
Swimming pools	Venues per 100,000; benefit; supply proxy	1

We use official statistical tables, operator inventories and municipal geodata to fill missing fields. For each admitted supplement, the archive records the provider, URL, year, geographic and population scope, units, extraction rule and file hash. Original PDFs and API responses are retained separately from derived tables. Search summaries locate evidence; numerical admission requires the underlying record or an explicitly identified provider capture. Values are converted to compatible units before utility calculation: for example, the supplied higher-education density per million is divided by ten to obtain the per-100,000 scale. Mode share is not converted into physical stop density.

4.2 Collection problems and their resolution

A station is not a platform, and a network is not a city. Oslo's municipal GIS represents platforms. Selecting records within the city boundary and consolidating names gives 89 metro stations from 186 platforms. Zurich's official 2025 VBZ report separately lists 446 stops on city territory and 235 in the surrounding area; only 446 enter the city inventory [5, 6]. A November 2026 snapshot of Helsinki's operator similarly distinguishes 19 metro stations in Helsinki from 11 in Espoo [7]. These are mode or operator subsets, so they supply lower bounds rather than complete all-operator totals. For a verified facility count c and fixed population P, density is $d = 10^5 c/P$; utility is bounded by $[d/(a+d), 1]$.

Madrid's metro register contains 293 records. Of these, 242 carry code 28079 for the city, independently confirmed by INE. Omitting two unnamed records without lines and consolidating 240 named records by the provider's network-station identifier yields 191 city complexes. Linked multi-name complexes are conservatively counted once; 51 outside-city records are excluded [23, 24]. This is a 29 May 2026 registered inventory with the supplied 2025 population, not a real-time operating count.

A transport register can include non-passenger objects. Vienna's versioned operator files group stopping positions into DIVA station areas. A broad route-pattern extraction yields 1,695 named city areas, but it includes depot objects whose passenger-stop status is unproved. We reject that count and retain 99 metro areas matched individually to the city's separate existing-station layer and administrative polygon [28, 29]. Explicit spelling aliases

are checked against both names and coordinates; 110 line appearances are not counted as 110 complexes. The 8 November 2026 export uses the unchanged supplied 2025 population. Its inventory membership does not guarantee a departure or unrestricted access at a particular time.

Prague's current city-classified PID list supplies 57 named metro complexes with route timetables [30]. Můstek, Muzeum and Florenc each count once despite two serving lines. The 2025 DPP report has 61 line appearances and 58 distinct complexes; the current subset omits Flora, whose 2026 closure is corroborated by the operator [31, 32]. Florenc's partial accessibility restriction does not close the whole complex. Both supplements retain upper utility 1 for uncounted facilities.

Berlin's nine official U-Bahn route profiles contain 131 station appearances in tariff zone A, defined by BVG as the Berlin inner city through the S-Bahn ring. Native station identifiers consolidate these to 100 complexes; normalized names are bijective with identifiers and cross-line transfers corroborate repeated appearances [33]. We retain this inner-city subset and leave outer stations and other modes uncounted. The November 2026 published inventory uses the supplied 2025 population; it measures registered resources rather than real-time service or a student's individual access.

Milan's municipal station layer supplies 130 records, of which 109 have both the Milano municipality name and code 015146. Consolidating the five line-specific interchange pairs identified on the operator map leaves 104 station/hub entities; 21 records without matching geography are omitted. The station layer is distinguished from its entrance layer, and proximity alone does not merge adjacent stations. Its stated inventory date is 16 October 2024; with the unchanged supplied 2025 population, it provides a historical lower-bound proxy rather than an exact 2025 count or current-service census [47].

London's TfL inventory contains 1,751 mixed objects, including platforms and entrances. Of 272 station-level records, 62 have the exact fare-zone value 1. Omitting two records conservatively and consolidating Paddington's shared native hub leaves 59 station/hub entities; all retained representative coordinates lie within the ONS December 2025 London-region boundary [36, 37]. This November 2026 inventory uses the unchanged supplied 2025 population. It raises the resource lower bound while leaving other zones and operators uncounted; it does not establish individual access.

A list of venues can contain closed facilities. Stockholm's current municipal swimming directory lists fourteen named venues, one of which is explicitly closed for renovation. The thirteen retained venues agree with the current overview; the temporary year-round replacement is counted once [8]. We count venues, not individual basins. This municipal subset gives a supply lower bound and leaves other providers unknown. Its access date is November 2026; a page-footer date is not used to reconstruct a 2025 inventory. Paris's 244 intra-muros metro stations come from a March 2022 atlas using RATP 2021 and Apur 2022 sources, not a 2025 observation [9]. A separate withdrawal restores this historical supplement to unknown.

Tokyo's retained municipal subset consists of five indoor swimming venues at different Nerima addresses, verified against original city pages with 25-metre specifications. Heiwadai's current closure with no reopening date excludes it; Miharadai's announced February–May 2027 closure is distinguished from current stock [25, 26]. Other wards/providers remain uncounted. The count measures venues, not lanes, basins or simultaneous opening.

Pupils and teachers must refer to the same schools and stage. Prague's 2024/25 tables report 80,073 secondary-school pupils and 6,877.1 teacher full-time equivalents (FTE),

including lower-gymnasium years. Subtracting the matching 8,748 pupils and 737.8 FTE yields 71,325 pupils and a rounding envelope of $[6139.2, 6139.4]$ FTE [10]. The Prague measure includes all study modes, including adult and non-daytime cohorts. Stockholm's published city-school ratio covers upper-secondary years 1–3 and all provider ownership types, measured on 15 October [11]. Its API value 16.42091405 is conservatively represented by $[16.35, 16.45]$, an intentional precision convention rather than a claimed measurement-error confidence interval. For Henri-IV (Paris, UAI 0750654D), 2025 records give 794 lycée pupils and 153.65 institution-level teacher FTE [34]. FTEs lack an upper-stage allocation; the 2026 directory flags post-baccalaureate provision without recovering 2025 workloads [35]. Public/contracted-private coverage also excludes other providers. These unmatched totals cannot identify Paris's city PTR, which remains unknown. Dublin's payroll quality report states that school-roll FTE are not published; the available county or national aggregates cannot establish a matching city denominator [27].

Rates require matching units and denominators. CBS's 2023 release gives ten provisional homicide victims in Amsterdam in 2022; its regional workbook instead totals 68 over 2018–2022. Neither matches the supplied numerator 340 [39]. Zurich's PKS report lists one completed offence in 2024 and two in 2025. Its glossary counts criminal acts independently of victims, and the submitted rate of 0.14 still lacks a verified denominator [40]. We retain both original entries and test their information contribution in Section 5.5, without constructing replacement rates across incompatible units.

Unavailable records and unobserved travellers are retained honestly. An initial Singapore census PDF endpoint returned 404; the official Parliament mirror and official data API supplied the report and matching table [13]. Duration bins are converted using utility at their endpoints, without bin midpoints or an artificial maximum duration for the open tail. The stage-specific total enrolment estimates exceed the travel-table estimates by estimated uncovered student totals of 1,811 for secondary, 989 for post-secondary non-tertiary and 1,635 for polytechnic. These are weighted survey estimates rather than complete headcounts. The uncovered mass retains utility $[0, 1]$; unknown stage proportions are handled by the interval convex hull. The resulting $[0.21048, 0.61884]$ is a 2020 school-stage proxy, not an identified 2025 interval for ages 14–18. When Dublin raw stop downloads returned 403, no count was inferred from the catalogue. A later official map supported a separately documented subset of 19 existing city-centre tram stops [14]. Warsaw's recovered 2022 mobility report gives a rounded mean school-trip duration of 27 minutes for a city subgroup with a stated sample base of 22. The survey covers residents aged 15 and above; the published school-travel chart does not isolate upper-secondary pupils or a matching 14–18 population [38]. Its travel utility remains unknown pending a matched school-stage denominator.

Named inventories make partial coverage reproducible. Singapore's LTA GeoJSON supplies 5,205 distinct nonempty bus-stop codes, without distance-based clustering or an independent operating-status census [41]. Stockholm's 2025 city–police report supplies thirteen metro stations within two city districts: eight in Bromma and five in Hässelby-Vällingby [46]. For swimming supply, a dated April 2025 ActiveSG article and linked venue records identify seven public Singapore venues [42]; Zurich's November 2026 directory identifies seven indoor venues [44]; Milan's tariff effective June 2025 names seventeen indoor venues across groups A and B [45]. Each pool inventory counts named physical venues once and retains its own date and coverage. The unchanged population denominators are retained, including Singapore's 2024 value; none of these inventories establishes a complete 2025 city census or continuous adolescent access.

Warsaw's yearbook supplies 119,740 pupils across six youth school routes in 2024/25, covering all ownership types by school location [43]. General-school teacher FTE include adult provision without a separate allocation. Allocating its unknown youth share over the full feasible range gives 2,490–8,144 FTE; an outward one-FTE guard for each of four published teacher cells gives 2,487–8,148. The guard represents a unit-rounding assumption. These bounds measure school resources and do not identify a resident age-14–18 population or a statistical confidence interval.

The current 160 cells comprise 112 point values, 34 partial intervals and 14 fully missing intervals: eight teacher-ratio, three transport and three travel entries. The original table had 112 points, 12 partial intervals and 36 fully missing cells. The twenty-two supplements therefore improve information without replacing the team's existing measurements. Facility supply, general commute proxies and school resources remain distinct from adolescent access, school travel and learning quality.

5 Establishing the baseline evaluation model

5.1 Determining indicator weights

We retain eight main indicators at weight 0.1125 each and two auxiliary facility indicators, transport stops and pools, at 0.05 each, so $\sum_j w_j = 1$. These weights express a transparent modelling choice. They are not estimated from a student survey. Equal weights provide a separate reference. TOPSIS is the primary score; VIKOR provides a focused method comparison. AHP represents explicit preference scenarios [2], and Monte Carlo measures sensitivity to declared distributions. These extensions retain the same criteria rather than introduce unrelated ranking models.

For preference multipliers $m_j > 0$, we use

$$w'_j = \frac{m_j w_j}{\sum_k w_k m_k}, \qquad A_{jk} = \frac{w'_j}{w'_k}. \tag{1}$$

The principal eigenvector of A recovers w'. With $\lambda_{\max}$ its largest eigenvalue, $CI = (\lambda_{\max} - N)/(N-1)$ and $CR = CI/RI_N$. Because this matrix is built from exact ratios, $\lambda_{\max} = N$ and $CR = 0$ up to numerical precision. This verifies internal consistency; it does not establish that these are students' actual preferences.

5.2 TOPSIS scoring calculation

We adapt TOPSIS [3] using fixed reference anchors to preserve a common scoring scale across evidence, preference and future scenarios. For benefit and cost measurements respectively,

$$u^+(x) = \frac{x}{a+x}, \qquad u^-(x) = \frac{a}{a+x}. \tag{2}$$

Each anchor in Table 1 is a declared reference scale, not an estimated optimum. Travel uses $q(t) = e^{-t/30}$ for duration t in minutes. When only mean duration $\bar{t}$ is known, Jensen's inequality gives $\mathbb{E}[q(t)] \in [e^{-\bar{t}/30}, 1]$. Fully missing utilities remain $[0, 1]$.

For utility vector u_i, weighted distances from fixed worst and best points are

$$D_i^- = \sqrt{\sum_j w_j u_{ij}^2}, \qquad D_i^+ = \sqrt{\sum_j w_j (1 - u_{ij})^2}, \qquad S_i = 100 \frac{D_i^-}{D_i^- + D_i^+}. \tag{3}$$

Weights enter the squared distances once. Since S_i increases with every utility, endpoints ℓ_i, h_i give $L_i = S(\ell_i)$ and $U_i = S(h_i)$. Under the retained rectangular uncertainty set, define

$$R_i = \max\{0, \max_{k \neq i} U_k - L_i\}, \qquad i^* \in \arg\min_i R_i, \tag{4}$$

$$\mathcal{C} = \{i : U_i \geq \max_{k \neq i} L_k\}. \tag{5}$$

Maximum regret R_i measures how far the selected city could trail another city. Set $\mathcal{C}$ retains every city that could still be first. The maximum excludes the selected city itself. These conditional data bounds are not sampling confidence intervals, and a conservative recommendation does not identify a uniquely best city.

5.3 Scoring results and interpretation

Zurich minimizes maximum regret, followed by Singapore and Oslo (Table 2). Zurich's score lower bound is below those of Singapore and Oslo; its own upper bound is excluded from its competitor set. Its regret is $70.90353 - 53.53548 = 17.36805$, whereas Oslo's is $72.18379 - 54.29594 = 17.88785$. All sixteen cities remain possible champions. The decision order expresses a conservative recommendation under the stated criteria and intervals.

Table 2: Conservative baseline decision ranks with unchanged weights.

Rank	City	Score interval	Maximum regret
1	Zurich	53.54–72.18	17.37
2	Singapore	54.33–63.53	17.85
3	Oslo	54.30–63.90	17.89
4	Stockholm	52.12–64.54	20.06
5	Prague	51.97–62.78	20.21
6	Dublin	51.13–70.90	21.05
7	Berlin	50.73–61.54	21.45
8	Helsinki	49.41–69.84	22.78
9	Paris	49.11–69.04	23.08
10	Madrid	49.00–68.22	23.18
11	Vienna	47.52–61.03	24.66
12	Warsaw	47.50–63.99	24.69
13	Milan	46.07–65.99	26.11
14	London	45.22–66.12	26.97
15	Tokyo	43.96–61.12	28.22
16	Amsterdam	42.29–60.43	29.90

Holding weights and formulas fixed, the twenty-two admitted supplements narrow the average fixed-weight score interval from 18.81077 to 16.06144, a 14.62% reduction. The Milan metro subset narrows the preceding fixed-weight mean from 16.08933 to 16.06144. The 104 retained station/hub entities raise one utility lower bound while all static decision ranks

remain unchanged. This is an information update using a dated inventory, not an observed improvement over time. Earlier additions improved Stockholm's decision position. Withdrawal of the historical Paris and Dublin transport inventories still recommends Zurich; withdrawal of Singapore's historical school-stage travel proxy also preserves this recommendation. Evidence improves precision and changes decisions before it necessarily reduces the possible-champion set.

As a method diagnostic, TOPSIS and VIKOR use the same illustrative midpoint utilities and weights. VIKOR balances total and largest weighted shortfalls equally and uses sample best/worst values [4]. TOPSIS selects Zurich at the midpoint, whereas VIKOR selects Dublin; Kendall's $\tau_b = 0.500$. Dublin satisfies VIKOR's advantage and stability conditions. Under equal weights, $\tau_b = 0.650$; Oslo has the lowest VIKOR index but fails those acceptance conditions, so it is not an accepted unique compromise solution. The midpoint comparison does not replace the interval decision in Equation 4. Agreement indicates similarity within the chosen conventions; disagreement exposes method dependence rather than justifying a post-hoc choice of the preferred winner.

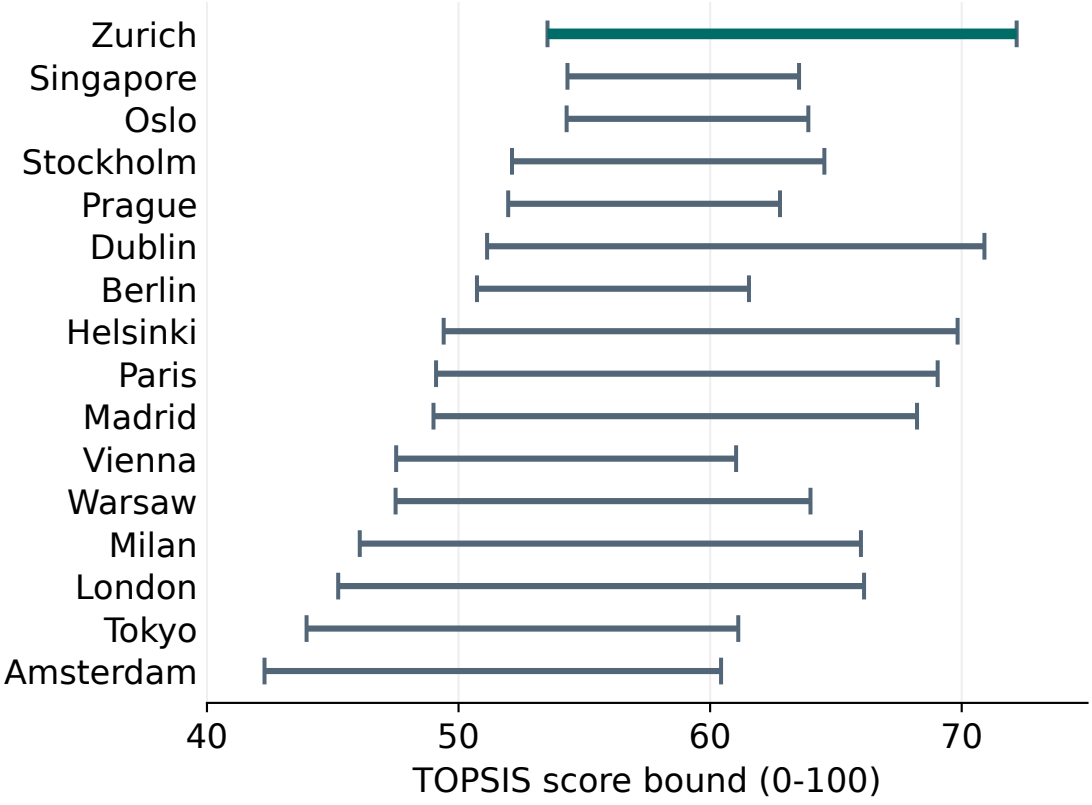


Figure 1: Current score bounds and conservative decision order for the sixteen cities. Intervals retain missing information; they are not confidence intervals. Source: frozen city matrix and admitted supplements.

5.4 A significant revision: retaining uncertain evidence

A controlled comparison shows why evidence bounds belong in the decision. On the same 160 utility entries, compare the midpoint completion $z_{ij} = (\ell_{ij} + h_{ij})/2$ with the retained-interval

formulation in Equation 4. Both use identical weights, anchors and TOPSIS scores. The midpoint is a declared diagnostic scenario, rather than an estimate of missing observations or a reconstruction of the team's historical first model.

The revision retains 48 non-point entries and selects the city with the smallest worst-case shortfall. Zurich remains the recommendation, but 13 cities change decision rank under fixed weights; Kendall's $\tau_b = 0.583$. With equal weights, 15 ranks change and $\tau_b = 0.650$.

For each target city c, construct a separate interior completion: set every uncertain utility to $\ell_{ij} + \alpha_i(h_{ij} - \ell_{ij})$, with $\alpha_c = 0.95$ and $\alpha_i = 0.05$ for $i \neq c$. All 112 point values remain fixed and all 48 uncertain values lie strictly inside their intervals. Each target ranks first under both baseline and equal weights. Across the sixteen targets, the smallest lead is 4.966 score points under baseline weights and 5.306 under equal weights. These separate completions establish feasibility in the declared utility rectangle; they carry no estimated probabilities and do not establish joint physical attainability.

Retaining intervals changes the decision order while preserving sixteen possible champions. The 14.62% reduction in interval width comes from admitted data, separately from this formulation comparison.

5.5 Sensitivity to the accepted input information

The conservative choice depends on which submitted observations are accepted. Of the 112 retained point entries, 64 are labelled 2026, two 2025 and 46 earlier; 28 carry non-`ok` source flags: 16 `MODELLED`, eight `STALE`, three `SUSPECT` and one `NATIONAL`. For a controlled test, replace selected point utilities by $[0, 1]$, keeping weights, all supplements and the 48 original non-point bounds unchanged. The groups are points dated before 2020, after 2025, non-`ok` points and their union. The union counts overlapping entries once.

Table 3: Recommendation after withdrawing selected submitted point information. Width is the mean baseline-weight score range on the 0–100 scale.

Withdrawn group	Cells	Baseline	Equal	Width
Retained input	0	Zurich	Zurich	16.06
Before 2020	13	Singapore	Zurich	21.14
After 2025	64	Zurich	Zurich	48.42
Non-ok flags	28	Berlin	Paris	29.11
Union	95	Singapore	Singapore	68.79

Withdrawing the 28 flagged points changes the recommendation to Berlin under baseline weights and Paris under equal weights. Removing the 13 pre-2020 points selects Singapore under baseline weights; equal weights retain Zurich. Withdrawing the 64 later-dated points retains Zurich under both references, while widening the baseline mean range from 16.06 to 48.42. The 95-entry union selects Singapore under both. All sixteen cities remain possible champions in every case. These withdrawals test dependence on accepted information: the labels and date cutoffs are not calibrated error probabilities or a judgment that every flagged entry is wrong. The experiment preserves the primary data and does not reconstruct a same-year-only matrix.

Withholding Zurich's homicide-rate point alone selects Singapore under baseline weights and Oslo under equal weights. Withholding Amsterdam's alone retains Zurich under both.

Each single-point test keeps the other 159 entries fixed; no official count is inserted as an assumed replacement rate.

Holding the other 159 entries fixed while varying this point rate x_Z, Zurich ties Singapore in maximum regret at about $x_Z = 0.23002$ per 100,000 under baseline weights, and Oslo at about $x_Z = 0.90473$ under equal weights. Below each respective cutoff Zurich uniquely minimizes regret; above it Zurich does not. These are hypothetical point-rate cutoffs, not observed rates or confidence limits.

6 Model revision and extension

6.1 Considering future changes

We examine three conditional changes while holding anchors and population denominators fixed. Resource expansion raises higher-education, stop and pool densities by 10%. Living pressure raises living costs, housing cost ratios and travel durations by 10%. Balanced improvement reduces pupil–teacher ratios, $PM_{2.5}$ and travel durations by 10%. A comparable multi-year panel is not available, so these changes have no estimated annual rate or forecast date.

For raw multiplier $r > 0$, the utility transformations are

$$T_r^+(u) = \frac{ru}{1 - u + ru}, \qquad T_r^-(u) = \frac{u}{u + r(1 - u)}. \tag{6}$$

These increasing maps propagate raw-scale changes through benefit and cost intervals. For known mean travel duration, the new Jensen interval is $[e^{-r\bar{t}/30}, 1]$. Singapore instead has bounds $[L_q, U_q]$ on expected utility. With $q = e^{-t/30}$ and $t' = rt$, Jensen and $0 \le q \le 1$ give

$$\mathbb{E}[q^r] \in \begin{cases} [L_q, U_q^r], & 0 < r < 1, \\ [L_q, U_q], & r = 1, \\ [L_q^r, U_q], & r > 1. \end{cases} \tag{7}$$

Raising both endpoints to r would generally misrepresent an expected utility. Equation 7 is an outer moment bound and need not be reversible. Fully missing travel remains unknown. Thus the extension accommodates the different information supplied by means and by travel distributions.

All three uniform-change scenarios retain Zurich as the conservative recommendation and all sixteen cities as possible champions under both weight references, including the Singapore-proxy withdrawal arm. Balanced improvement changes the fixed-weight second and third decisions from Singapore–Oslo to Oslo–Singapore. We also solve a decision boundary: holding other cities fixed, proportionally reduce one city's pupil–teacher ratio, $PM_{2.5}$ and travel durations until its maximum regret ties that of the best other city. Oslo reaches the tie at approximately 5.685% reduction, Singapore at 10.309%, and Stockholm at 28.875%. Zurich already minimizes regret at baseline. A slightly greater reduction produces a strict recommendation switch in these calculations. These thresholds describe the model response to a specified joint change; they do not establish policy feasibility or a uniquely identified champion. Berlin's hypothetical combined-cost tie threshold is 38.084% after the Milan inventory update. Scenario definitions and weights are unchanged.

For a separate future-shock simulation, each draw combines a common shock and city-specific shocks under a declared bounded distribution, with the criterion weights held fixed. Original data intervals are not randomly completed. Specifically, $C \sim U[0,1]$ is shared across cities and $I_i \sim U[0,1]$ varies by city, independently; $\delta_i = 0.05C + 0.05I_i$. The relevant raw multipliers are $1 + \delta_i$ or $1 - \delta_i$, and affected criteria within a city share that multiplier. With 10,000 draws per scenario, Zurich's fixed-weight recommendation frequencies are 100.00%, 96.85%, 97.72% for resource expansion, living pressure and balanced improvement respectively. Repeating the same shocks after withdrawing Singapore's historical stage-travel proxy gives 100.00%, 99.19%, 97.72% respectively. These paired calculations isolate dependence on that proxy while retaining the chosen shock distribution. These are conditional stress frequencies under an uncalibrated distribution. Weight perturbations below answer a separate question and are not blended silently into the future-shock frequencies.

6.2 Considering different preferences and development situations

Education priority doubles the weights of the two education indicators. The independent living scenario doubles travel and transport-stop weights. Equation 1 then normalizes all weights. Education priority changes the conservative recommendation from Zurich to Oslo, with decision-rank $\tau_b = 0.733$ relative to baseline. Independent-living priority retains Zurich and gives $\tau_b = 0.883$. All sixteen cities remain possible champions in each profile. At concentration 100 around baseline weights, Milan's mean decision rank changes from 13.3073 to 12.9648 after its inventory supplement. The fixed education-priority recommendation remains Oslo; these are changes in the declared evidence bounds and weight distribution. The resulting differences show how the definition of a desirable city affects the decision. They do not establish measured differences between fourteen- and eighteen-year-old students.

6.3 Perturbing weights

For each preference centre w', we draw $w^{(b)} \sim \text{Dirichlet}(\kappa w')$, with $\kappa \in \{20, 100, 500\}$ and 10,000 draws per setting. Every draw recomputes interval TOPSIS and maximum regret without filling missing values. We report mean decision rank, fractional first-place recommendation frequency and fractional top-three decision membership; ties share membership.

Table 4: Conditional recommendation frequencies (%) for selected cities at concentration 100.

Preference centre	Zurich	Oslo	Singapore
Baseline	43.33	27.92	28.23
Education priority	6.83	42.95	41.43
Independent living	58.99	15.46	25.12

At concentration 100, Oslo has a better mean decision rank (2.1092) than Zurich (2.2620), although Zurich has the greater recommendation frequency and is the fixed-weight minimax choice. These summaries answer different questions about the chosen weight distribution. These two summaries optimize different quantities. We retain minimum maximum regret as the predeclared decision rule; Monte Carlo reports its sensitivity. Neither average ranks nor a network aggregation resolves the missing empirical information. A low AHP consis-

tency ratio and repeated simulations demonstrate calculation properties, not representative student opinion.

6.4 Within-city fairness and growth stages

Citywide facilities per resident measure supply, not who can use it. Fees, opening hours, transport links, disability access and neighbourhood location can separate nominal supply from student opportunity. Without matched student locations and qualifying facilities, the city-level matrix cannot identify these access distributions.

A separate Shanghai school-resource case uses 16 districts, 235,542 regular senior-secondary pupils and 23,999 full-time teachers in the 2025 school year. The pooled utility is 0.604481; the student-weighted mean of district utilities is 0.603267. Ordering districts by resource utility and averaging the bottom 20% of pupil mass, with fractional inclusion of the boundary district, gives 0.580954, 0.022313 below the weighted mean. These are district-homogeneous school-location resource scenarios, not observed individual teacher exposure or resident accessibility [15]. To extend the same education criterion to districts d, let n_d be the matched pupil count and r_d its pupil–teacher ratio. The student-weighted utility is $\bar{u} = \sum_d n_d \, 15/(15 + r_d) / \sum_d n_d$. Its difference from a citywide ratio reflects spatial aggregation, not an effect of any particular policy. A school-location analysis also does not locate resident students or establish their travel access.

For future access measurement, define a_s as a student's qualifying access measure, incorporating the relevant eligibility, price and time conditions. With student weights p_s, report the weighted mean and the bottom-20% mean using fractional inclusion at the threshold. A common safety or travel standard could also be reported as weighted coverage. These require additional matched data; substituting a municipal average would not measure fairness.

A historical Istanbul household survey offers current-high-school groups aged 14–15 and 16–18. Their travel-utility marginal bounds overlap: $[0.35961, 0.55711]$ and $[0.29347, 0.59346]$, respectively. They include unknown durations and optional members; shared unknown-age mass means their extrema need not be jointly attainable. They describe 2012 broad groups, not exact ages fourteen and eighteen today. Singapore's MOE table records 38,555 students aged fourteen and only 165 aged eighteen at the start of 2020 still in secondary schooling; the latter is not an all-eighteen-year-old resident denominator. These counts cannot be joined to census school-stage travel bins without a matching population/reference-day table [16, 17].

The shared-membership constraint permits both historical age-group orderings. For younger-minus-older mean travel utility, the joint closure is $[-0.23330, 0.26325]$, assigning the shared unknown-age mass once. Two separate completions with unknown utilities 0.05 and 0.95 give contrasts $+0.23842$ and -0.20848. Both use finite durations and allocate the shared mass to the younger group. The closure endpoints allow zero utility as a limiting value; the two reversals use strictly interior values. These are feasibility constructions within the retained 2012 aggregate uncertainty set, not observed age effects.

Age can affect observed travel, permissions and opportunities separately from preference weights. Comparing exact ages fourteen and eighteen requires comparable resident utilities $u_{ij}^{(14)}, u_{ij}^{(18)}$ and independently elicited weights $w_j^{(14)}, w_j^{(18)}$. Our analyst preference scenarios carry no age labels; matched student data are required to compare access.

7 Comparing cities within the same country

Every primary city belongs to a different country, so the main sample cannot identify within-country city contrasts. We use the existing United States and India cases as frozen legacy25 V117 examples with the same ten indicator definitions and scoring formula [20]. They do not enter the sixteen-city ranking, and comparison within a country does not remove differences in source year, city boundary or educational coverage.

Table 5: Separate historical country cases using the legacy 25-city V117 evidence.

Country	City	Score interval
United States	Phoenix	31.39–71.68
	New York	30.22–72.81
India	Mumbai	28.82–60.64
	Delhi	32.59–63.90

Phoenix has higher retained affordability utilities than New York. Mumbai has the higher $PM_{2.5}$ utility, while Delhi has higher living-cost, housing, school-resource and broadband utilities in the retained evidence. The cases inherit the historical study's geographic and educational scope; their score pairs overlap, so neither case identifies a necessary overall winner. Affordability contrasts use Numbeo's 2025 named-city aggregates [21]; the Mumbai–Delhi air contrast uses 2021 WHO monitoring means [22]. These source scopes are retained in the case source index. These descriptive contrasts do not identify a national causal effect.

8 Conclusion and decision use

Under the retained team inputs, Zurich minimizes maximum regret with baseline and equal weights, while all sixteen cities remain possible champions. The recommendation gives a family a defensible starting comparison under explicit criteria and source bounds. Education priority changes that starting choice to Oslo, demonstrating a practical reason to discuss priorities before using a city order. Withholding Zurich's unverified homicide-rate point alone selects Singapore under baseline weights and Oslo under equal weights. The wider 28-point source-flag withdrawal selects Berlin and Paris, respectively. Choosing a city therefore requires both an explicit preference rule and scrutiny of the accepted sources.

Verified supplements reduce average score-interval width by 14.62% without replacing the original table's values. The controlled formulation comparison then holds that evidence fixed: retaining uncertainty changes 13 decision ranks and preserves completions in which another city exceeds Zurich. These are distinct improvements in information and interpretation, rather than a claim that the model has measured a city's true superiority.

A city choice built around a student's day

Under the retained team inputs, Zurich is the cautious recommendation emerging from our comparison of sixteen cities for high-school life. An alternative that places greater emphasis on education points to Oslo. Together, these findings offer students and families a practical starting point: a city comparison that connects everyday opportunities with personal priorities while keeping gaps in the evidence visible.

For a teenager, choosing a city involves the school day and the hours around it. Teaching resources, nearby higher education, clean air, safety, housing and living costs, transport, journey times, internet access and swimming facilities can all matter. Our study brings these ten considerations into one framework. Its purpose is to help families examine trade-offs, rather than assume that the same place will suit every student.

A cautious recommendation with a clear meaning

The comparison uses common scoring yardsticks and a stated balance of priorities. Where a figure is missing or describes part of a city's resources, the model keeps a range of possible scores. An unknown figure therefore remains open rather than being replaced with an invented average. Each city's overall result is a range, reflecting the information available under the model's assumptions.

The recommendation asks how far a selected city could fall behind a better-scoring alternative within those ranges. Zurich has the smallest worst-case shortfall under the baseline priorities. It also remains the cautious recommendation when all ten considerations receive equal weight. This makes it a useful place to begin a broad comparison; all sixteen cities nevertheless remain possible leaders within the retained evidence ranges. The study has not established a single certain winner.

Better evidence makes the comparison more useful

Official records become useful when their boundaries and units are clear. Vienna's broad transport register contains depot objects whose status as passenger stops is unproved. Cross-checking the operator list with a separate city map confirms 99 metro station areas. Prague's city list supplies 57 metro complexes after interchanges are counted once and a closed station is omitted. These examples show why a useful inventory needs more than a headline total.

Across twenty-two previously missing entries, additional evidence reduced the average width of the city score ranges by about 15 per cent. The original submitted values and scoring rules stayed in place. Narrower ranges make the comparison more informative, while the full set of possible leaders shows that further evidence could still change the conclusion. The records span different years, and some describe school or city resources rather than the experience of a particular teenager.

Students' priorities can change the recommendation

In an education-focused comparison, the relative emphasis on teaching resources and the density of higher-education institutions is doubled before the priorities are balanced again. Oslo then becomes the cautious recommendation. This is an illustrative choice made by the analysts, not a finding from a student survey. It gives a family a concrete way to explore how a stronger educational emphasis changes its shortlist.

City totals also need a local check. A transport-stop inventory leaves questions about walking distance, service frequency, fares and accessible routes. A pool count leaves questions about entry fees, opening hours and which neighbourhoods the facilities serve. These are useful questions for families and city planners to investigate before treating resource supply as equal access. The study does not measure an individual student's benefit from adding a stop, teacher or pool.

Use future scenarios to ask better questions

The study also examines three hypothetical changes: expanding selected resources, increasing living and travel pressures, and reducing selected educational, air-quality and travel burdens. When the affected quantities change uniformly by ten per cent, Zurich remains the cautious recommendation under both baseline and equal priorities. Repeated simulations explore what happens when the assumed changes differ across cities. They test chosen scenarios; they do not predict a future year or estimate a city's real chance of being best.

Checking a specific source can change the shortlist: treating Zurich's unverified homicide-rate figure as unknown points to Singapore under the original priorities and Oslo with equal priorities. The original figure remains in the declared baseline; no replacement rate is assumed.

A family's next step is to pair the city comparison with its own school options, budget, daily route and preferred activities. Students can give education more emphasis, examine the affordability trade-offs, and check whether nearby opportunities fit their routines. The model makes those choices easier to discuss: a clear starting recommendation, room for different priorities, and an explicit view of the evidence needed for a more personal decision.

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Report on Use of AI Tools

Documentation status: incomplete review record. This appendix records assistance supported by saved task metadata and research receipts. The complete team AI history remains to be reconciled with actual saved conversations and other tool records before submission.

Known tool and scope

OpenAI Codex, used 8–9 November 2026. The task log records Codex Desktop client version `0.166.0-alpha.5.1` and model identifier `gpt-6.1-sol` in all 28 saved turn contexts within the inspected prefix. The identifier does not expose an immutable serving build or establish the model used by every concurrent assistant. The assistant supported official-source search and extraction, data-scope checks, Python calculations and audits, LaTeX drafting, and wording revisions. Concurrent assistant tasks prepared bounded evidence checks, a controlled model diagnostic and the news-article contribution [18]. The anti-defensive-writing skill supplied editorial instructions; it is a workflow resource, not a surveyed student preference or a data source.

Exa, used 8–9 November 2026 through the recorded `web_fetch_exa` tool, retrieved and extracted text from official webpages, PDFs and data interfaces [19]. Saved receipts identify the service, source URLs and returned text, but do not specify an internal model or service version. They do not establish whether a generative model participated in every extraction. Extracted representations are distinguished from original downloaded bytes. For example, Tokyo sports-facility PDF text with ambiguous row order was excluded; the admitted five-venue lower bound uses separately downloaded Nerima municipal HTML.

Recovered records and coverage

A complete-line prefix of the local task log is authenticated by its byte count and SHA-256 in the supporting metadata inventory. That prefix ends at 03:17 UTC on 9 November 2026; later continuations are outside its coverage. A private export preserves four genuine request messages, 99 visible progress messages and 22 complete visible final messages, with per-record hashes. It excludes reasoning, internal instructions, injected continuations and tool payloads. Consequently it does not cover generated code or manuscript text contained in tool calls, earlier tasks, attachments or other team members' use. The raw local log and private export are not included as submission transcripts. No missing prompt or output has been reconstructed and labelled as an exact historical quotation.

Verification associated with this draft

The research calculations preserve the original city CSV and the V154 primary evidence. The V154 validation independently recomputes weighted norms and self-excluded competitor maxima, verifies interval membership of constructed completions and checks source hashes. Original provider records are kept separately from derived tables. Compilation and rendered-page review check the presentation. These checks establish the specified calculation and document properties; they do not replace the team's responsibility to review factual claims, original supplied data and final authorship.

Records required to finalize this report

The final report must identify every AI tool actually used, its available model or version, date and purpose, and associate relevant use with the text and references. Actual saved query and response records should be inserted according to the applicable COMAP policy, including its stated exceptions. Other team members' tools, translation use and earlier conversations remain to be confirmed from their records.

The current in-text disclosure and references identify Codex and Exa assistance. Submission readiness also requires a verified Team Control Number in place of the review header and confirmation that this report accounts for the team's actual use.